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Start Here: Project Map

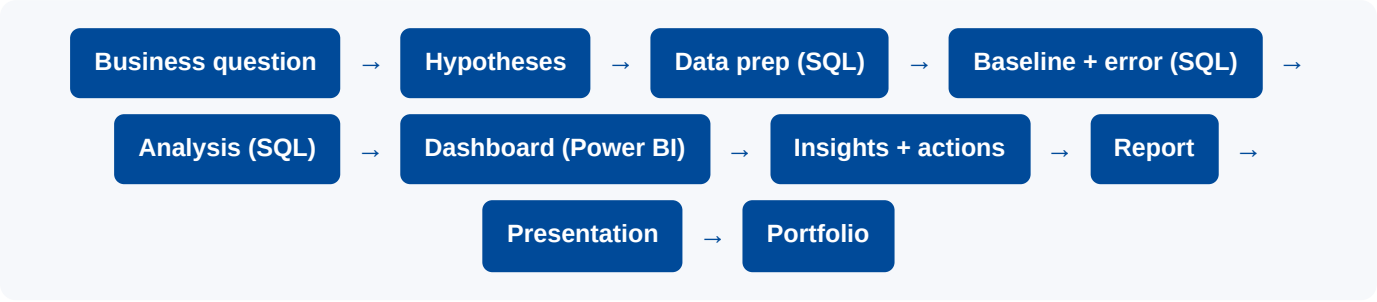
> **In simple words**
>
> This is the map of the whole project. It shows the order of work and which document helps at each step.

What this project is

RetailPulse is a grocery retail team. We study daily sales from the Kaggle **Store Sales - Time Series Forecasting** data. We build a **simple baseline forecast** in SQL, measure its error, and explain **where, when and why** demand is hard to predict. Then we tell the business what to do about it.

> **Your lane: Insights Analyst**
>
> You are the person who talks to the business, finds the story in the data, and helps people decide. SQL and Power BI are your tools. **The recommendation is the product.**

Project flow



Which document to use, and when

Doc	Name	Use it when	You produce
01	Project Charter	Day 1: define the problem and the decisions	Problem statement
02	Stakeholder Needs	Day 1: gather demand (requirements)	Requirement log
03	Data Dictionary and Model	Before writing SQL	Clean tables, holiday flag
04	Error Metric Framework	Before calculating error	Metric views
05	SQL Analysis Specification	While doing analysis	12 analysis outputs
06	Power BI Requirements	While building the dashboard	5-page dashboard
07	Python (Optional)	Only for quick charts	2-3 charts
08	Report Guidelines	While writing the report	Final report
09	Plan and QA Checklist	Daily, and before sharing	Checked work
10	Analyst Guide	When you feel lost	Clear thinking
11	Fast Track Steps	Every day: what to do next	Finished project
12	Hypothesis and Insight Playbook	While turning numbers into insights	Insight log
13	Executive Summary and Presentation	Last stage	Summary, slides
14	Interview and Portfolio Guide	After the project	Resume, README, story

Five rules of an Insights project

1. Every number must answer a business question.
2. Every finding needs a **so what?** and a recommended action.
3. Always compare. A number alone means nothing. Compare with another group, period or the overall value.
4. **Never invent numbers.** Our data has no price or cost. Measure impact in units, not money.
5. Be honest about limits. Say what the data cannot tell us.

Folder structure

```
D:\Project\Demand Forecasting Analysis\
  00_Project_Documents\        (this pack)
  01_Datasets\raw\             (Kaggle CSVs - never edit)
  01_Datasets\processed\       (SQL outputs)
  02_SQL_Scripts\              (numbered .sql files)
  03_PowerBI\                  (.pbix)
  04_Report\                   (final report)
  05_Presentation\             (slides)
  06_Python_Optional\          (quick charts)
  07_Screenshots\              (for README)
  README.md
```

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Project Charter and Business Problem

> **In simple words**
>
> This document explains why we do this project. We find where demand is hard to predict, so the business can make better stock decisions.

1. Business context

RetailPulse runs grocery stores in many cities. Every day, each store must decide how much of each product family to stock. A **forecast** is a guess of future sales. When the guess is too high, stock is wasted or sits unsold. When it is too low, shelves are empty and customers leave.

2. Problem statement

> **The decision problem**
>
> Supply Chain and Category teams need to know **where demand is hard to predict**, so they can decide where to add safety stock, where to review orders, and how to plan for holidays and promotions. Today they have no clear view of where forecasts miss, by how much, and in which direction.

> **Important assumption**
>
> The Kaggle data has no company forecast. We build a **simple baseline forecast** in SQL (same weekday last week). So our results show **where a simple baseline fails**, which means where demand is hard to predict. They are not the errors of an existing company forecast.

3. Business questions we must answer

#	Question	Type
A	Which product families are hardest to predict?	Where (category)
B	Which states, cities and store types are hardest?	Where (region)
C	When does error rise: months, holidays, paydays, promotions?	When (season)
D	Do we guess too high (excess stock) or too low (empty shelves)?	Direction
E	What should the business do about it?	Action

4. Decisions this project supports

Decision	Who decides	Evidence we give
Which families need a better forecast method first	Supply Chain Manager	Top high-error families and Pareto
Where to add safety stock	Category Manager	Under-forecast (shortfall) units by family and month
Where to cut orders to reduce excess	Supply Chain Manager	Over-forecast (excess) units by family and region
How to plan for holidays and promotions	Planning Team	Error on holiday, payday and promotion days
Which regions need support	Regional Manager	WAPE and Bias by state, city, store type

5. Scope

Included	Not included
Historical sales analysis (Kaggle train, stores, holidays, oil, transactions)	Machine learning or advanced forecasting models
Simple SQL baseline forecasts and error measurement	Changing the company buying process
Hypotheses, insights and recommendations	Money impact (no price or cost data)
Power BI dashboard, report, presentation	Real-time data pipelines

6. Success criteria

- At least 6 insights, each with evidence, a comparison, a reason, and a recommended action.
- Top high-error families named and ranked by share of total error.
- A dashboard where a manager finds the main message in 30 seconds.
- A one-page executive summary and a final report.
- All dashboard totals match the SQL totals.

7. Limits we state openly

- No price, cost or stock data, so impact is measured in **units**.
- The baseline is simple on purpose. Better methods may do better.
- Reasons for error are **likely causes** unless the data proves them.

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Stakeholder Needs and Demand Gathering

> **In simple words**
>
> Demand gathering means finding out what the business really needs before we start. We list who needs what and which decision each need supports.

1. Who cares about this project

A **stakeholder** is a person who uses or is affected by your work. This is a portfolio project, so the interviews are **simulated**. Write the answers as clearly marked assumptions.

Role	Main worry	Decision they make	What they need from us
Supply Chain Manager	Wrong order quantities	How much to order	Ranked list of high-error families, over and under direction
Category Manager	Empty shelves in key families	Safety stock by family	Shortfall units by family and month
Regional Manager	Some cities look worse	Where to give support	WAPE and Bias by state, city, store type
Planning Team	Holidays and promotions break the plan	Event planning	Error on holiday, payday, promotion days
Finance Manager	Cost of excess stock	Budget for stock	Excess units by family (money needs cost data)
You (Insights Analyst)	Giving advice nobody can use	What to recommend	Clean data and clear metric rules

2. Interview guide (8 questions)

#	Ask	Why we ask
1	What decision will you make with this analysis?	Links every result to an action
2	Which product families hurt most when forecast is wrong?	Sets priority
3	Is too much stock or empty shelves the bigger problem?	Decides which direction matters more
4	What error level is acceptable for a normal family?	Helps set the high-error rule
5	Which events change sales in your cities (holidays, paydays, local fairs)?	Gives hypotheses to test
6	How do you review forecast accuracy today?	Shows current pain
7	How often do you need the dashboard updated?	Sets refresh needs
8	What would make you trust this dashboard?	Sets quality rules

3. Requirements (what we must deliver)

ID	Requirement	Priority
R1	Overall WAPE, Accuracy %, Bias %	Must
R2	Ranking of families by WAPE and by share of total error	Must
R3	Over-forecast and under-forecast units by family	Must
R4	Error by state, city and store type	Must
R5	Error by month, holiday, payday, promotion	Must
R6	Written recommendations with owner and priority	Must
R7	Drill-through: daily actual vs forecast for one family	Should
R8	Comparison of two simple baselines by family	Should
R9	Oil price and transactions context	Could

4. KPI definitions agreed with stakeholders

KPI	Plain meaning	Defined in
WAPE	Total error divided by total actual sales	Doc 04
Accuracy %	1 minus WAPE	Doc 04
Bias %	Are we too high (+) or too low (-) overall	Doc 04
Excess units	Units forecast above actual	Doc 04
Shortfall units	Units actual above forecast	Doc 04
Error share %	How much of total error one group causes	Doc 04

5. Requirement log (fill this in)

ID	Stakeholder	Need (in their words)	Decision it supports	Priority	Where answered

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Data Dictionary and Data Model

> **In simple words**
>
> This document is a map of our data: what each column means, how tables join, and the traps to avoid (holiday double counting, missing dates, stores not open yet).

1. Tables we use

Table	Grain (one row is...)	Use
train	one store, one family, one date	Actual sales (main fact)
stores	one store	City, state, type, cluster
holidays_events	one holiday or event record	Holiday and event flags
oil	one date	Optional context
transactions	one store, one date	Optional context

We do **not** use `test.csv`. It has no sales, so we cannot measure error on it.

2. Column meanings

train

Column	Meaning	Note
id	Row ID	
date	Sales date	Daily data. Some dates are missing (for example 25 December)
store_nbr	Store ID	Links to stores
family	Product category	Our category field
sales	Units sold	Can have decimals (items sold by weight)
onpromotion	Items on promotion that day	A count, not only 0 or 1

stores

Column	Meaning	Example
store_nbr	Store ID	1, 2, 3
city	City	Quito
state	State	Pichincha
type	Store type	A, B, C, D, E
cluster	Group of similar stores	1 to 17

holidays_events

Column	Meaning	Values
date	Date of the record	
type	Kind of record	Holiday, Event, Transfer, Bridge, Additional, Work Day
locale	Scope	National, Regional, Local
locale_name	Region or city name	Ecuador, Pichincha, Quito
description	Name	Christmas
transferred	Holiday moved to another day	True / False

3. Things to know before you start

- Public-sector wages are paid on the **15th and the last day** of each month. This can change sales (payday effect).
- A big earthquake hit Ecuador in **April 2016**. It appears in the events data. Treat it as a special event.
- Some stores opened late. Some families have zero sales for long periods. Both can distort the baseline.
- After loading, check counts with the Kaggle page: about 3 million rows in train, **54 stores**, **33 families**.

4. Holiday join: how to avoid double counting

> **The risk**
>
> One date can have more than one holiday record. Joining train to holidays_events on `date` alone repeats sales rows and **double counts sales**.

1. Remove rows where `transferred = True`. That date was not really celebrated.
2. Match by scope: **National** = all stores. **Regional** = `stores.state = locale_name`. **Local** = `stores.city = locale_name`.
3. Build a **store-date holiday flag table** with exactly one row per `store_nbr` and `date`: columns store_nbr, date, is_holiday. Use DISTINCT or EXISTS.
4. Join train to this flag table, never to holidays_events directly.

is_holiday = 1 only for types Holiday, Transfer, Additional and Bridge. **Event** is kept separate (for example Black Friday). **Work Day** is a normal day.

5. Grain of the error table

Grain means what one row stands for. Our error table has one row per `store_nbr` x `family` x `date`. We add flags to it: is_holiday, is_payday, onpromotion, month, quarter, season. All WAPE numbers come from summing these rows.

6. Data model for Power BI (star schema)

```
dim_date      dim_store      dim_family
    \            |            /
     \           |           /
          fact_error
  (one row per store x family x date)
  flags inside the fact: is_holiday, is_payday
```

Fact columns: date, store_nbr, family, actual_sales, forecast_sales, abs_error, excess_units, shortfall_units, onpromotion, is_holiday, is_payday. Dimensions give names and groups for slicers.

7. Data cleaning rules (done in SQL)

Check	Action
NULL in sales or date	Count them. Remove or explain.
Duplicate rows (same store, family, date)	Check count. Remove if found.
Missing dates	Expected for 25 December. Do not fill. Make sure baseline does not use a wrong day.
Rows before a store's first sale date	Exclude. The store may not be open yet.
Families with zero sales for long periods	Mark them. Exclude or flag in high-error rule.

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Forecast Error Metric Framework

> **In simple words**
>
> This document defines the math rules for error: what the baseline is, how WAPE, Bias and units are calculated, and when a family counts as high error.

1. Baseline forecasts (simple on purpose)

Baseline	Rule	Role
A: Same weekday last week	Forecast for a day = actual sales 7 days before (same store, same family)	Main baseline
B: 4-week same-weekday average	Average of actual sales 7, 14, 21 and 28 days before	Comparison baseline

Both use only past data. The first 28 days have no baseline and are excluded. Always check that the earlier date is exactly 7 (14, 21, 28) days before, because some dates are missing.

2. Metrics

Metric	Formula	Plain meaning
Absolute error	ABS(actual - forecast)	Size of the miss on one row
WAPE (main metric)	SUM(abs error) / SUM(actual)	Error as a share of sales. Works with zero-sales rows.
Accuracy %	1 - WAPE	How close we are
Bias %	(SUM forecast - SUM actual) / SUM actual	Plus = guessing too high. Minus = too low.
Excess units	SUM of MAX(forecast - actual, 0)	Units we forecast but did not sell
Shortfall units	SUM of MAX(actual - forecast, 0)	Units sold above forecast (risk of empty shelf)
Error share %	Group abs error / total abs error	How much of the total problem this group causes
MAE (optional)	AVERAGE of abs error	Average miss in units
MAPE (optional)	Average of abs error / actual	Fails when actual = 0. Avoid as main metric.

> **Never average WAPE values**
> WAPE for any group = SUM of absolute error for that group / SUM of actual sales for that group. Averaging WAPE values from smaller groups gives a wrong answer.

3. Worked example (example numbers only)

Month	Abs error	Actual sales	WAPE
Month 1	100	500	20%
Month 2	200	1,000	20%
Month 3	300	500	60%
Total	**600**	**2,000**	**30%**

The correct total WAPE is 600 / 2,000 = 30%. The average of 20%, 20% and 60% is 33.3%, which is wrong.

4. Why we show both error and bias

A family can have a small Bias % but a big WAPE. This happens when over-forecasts and under-forecasts cancel each other. Bias hides the problem. WAPE shows the size of the miss. Bias shows the direction. Always report both.

5. High-error rule

> **Starting rule**
>
> A family is **High Error** if its WAPE is above **30%** and its total sales are above a minimum volume. Rank families by **Error share %** to find the biggest problems.

- 30% is a starting point. After the first run, look at the WAPE spread and adjust.
- Ignore tiny-volume families when flagging, because WAPE is unstable for them. Report them separately.
- Priority = high WAPE **and** high Error share. A high-WAPE family with tiny sales matters less.

6. Reporting levels

Level	Used for
Overall	Executive KPIs
Family	Category ranking, Pareto
State, city, store type, cluster	Region analysis
Month, quarter, named period, weekday	Season analysis
Holiday, payday, promotion flag	Event analysis

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SQL Analysis Specification

> **In simple words**
>
> This is the to-do list for SQL. Each question has a business purpose, a hypothesis to test, the SQL plan, and the insight to look for.

1. SQL rules for this project (PostgreSQL)

- One numbered `.sql` file per step. Add a comment header: purpose, input, output.
- Build in layers using views: clean data, then holiday flags, then baseline, then error table, then analysis.
- Use CTEs for readability. Use window functions for the baseline and for ranking.
- Use `EXTRACT(MONTH FROM date)` for months. Use `NULLIF` to avoid divide by zero.
- Check every result: totals must match across queries.

> **Four traps to avoid**
>
> **1. Holiday join:** use the store-date flag table (Doc 03).
> **2. Missing dates:** check that the earlier date is exactly 7 days before, do not trust row order.
> **3. Store not open yet:** exclude rows before a store's first sale date.
> **4. Zero sales:** use WAPE, not MAPE.

2. Build steps before analysis

File	What it builds	Check
01_create_tables.sql	Tables for the 5 CSVs	Columns and types match Doc 03
02_load_data.sql	Load CSVs	Row counts match Kaggle
03_data_checks.sql	NULLs, duplicates, missing dates, store start dates	Issues listed and handled
04_holiday_flag_table.sql	One row per store and date with is_holiday	Row count = stores x dates, no duplicates
05_baseline_forecast.sql	Baseline A and B using LAG / window functions	Earlier date is exactly 7 days before
06_error_table.sql	Fact table: actual, forecast, abs error, excess, shortfall, flags	Total sales equals train total after exclusions

3. Analysis questions

Each question has a **hypothesis** (our starting guess). We write it before running SQL, then test it. A hypothesis can be wrong. That is also an insight.

Q1. How healthy is the baseline overall?	
Business question	Overall, how accurate is the baseline, and is it getting better or worse over the years?
Hypothesis	Overall WAPE is moderate, and it does not change much between years.
Method	SUM abs error / SUM actual overall and by year. Add Bias %, excess and shortfall units.
Output	`overall_health.csv` : year, WAPE, Accuracy, Bias %, excess units, shortfall units.
Insight to look for	Is the problem everywhere or only in some places? This sets up the next questions.

Q2. Which families are hardest to predict?	
Business question	Which families have the highest WAPE, ignoring tiny-volume families?
Hypothesis	Low-volume and special families (for example books, baby care) have the highest WAPE.
Method	WAPE by family. Add total sales. Apply a minimum volume rule. RANK by WAPE.
Output	`family_error_summary.csv` : family, total sales, WAPE, Accuracy, Bias %, high_error_flag.
Insight to look for	Are high-error families big or small in sales? Big ones matter more.

Q3. Which families cause most of the error? (Pareto)	
Business question	Which few families make up 80% of total error?
Hypothesis	A small group of high-volume families causes most of the error units.
Method	Abs error by family, share of total, `SUM() OVER (ORDER BY ... )` for cumulative share, mark the 80% cut.
Output	`family_pareto.csv` : family, abs error, error share %, cumulative share %.
Insight to look for	The shortlist to fix first. This is usually your strongest recommendation.

Q4. Are we too high or too low? (Direction)	
Business question	For each family, do we mostly over-forecast (excess) or under-forecast (shortfall)?
Hypothesis	Some families lean to excess and others to shortfall. The pattern is not the same for all.
Method	Excess units and shortfall units by family. Bias % by family. Label as Over, Under or Balanced.
Output	`family_direction.csv` : family, excess units, shortfall units, Bias %, direction label.
Insight to look for	Different actions: cut orders where excess is high, add buffer where shortfall is high.

Q5. Which regions are hardest?	
Business question	Which states, cities, store types and clusters have the highest WAPE and Bias?
Hypothesis	Smaller cities and some store types have higher WAPE because volumes are lower.
Method	Join error table to stores. WAPE, Bias % and total sales by state, city, type, cluster. RANK.
Output	`region_performance.csv`: state, city, type, cluster, sales, WAPE, Bias %, rank.
Insight to look for	Is error linked to size (low volume) or to location? Compare WAPE with sales.

Q6. Which months and seasons are hardest?	
Business question	How does error change by month, quarter and named periods (Christmas, Carnival, school start, Mother's Day)?
Hypothesis	December has the highest error because of Christmas demand.
Method	WAPE and Bias % by month and quarter. CASE WHEN to label named periods. Compare each with the overall WAPE.
Output	`season_error.csv`: month, quarter, period label, WAPE, Bias %, sales.
Insight to look for	Which period needs extra planning, and does it lean to excess or shortfall?

Q7. What is the effect of holidays and events?	
Business question	Is error higher on holidays, and one week after a holiday?
Hypothesis	Error is higher on holidays. Exactly one week after a holiday, the forecast is too high (excess), because the baseline copies the holiday spike.
Method	Use the holiday flag table. WAPE and Bias % for holiday vs regular days. Add a flag for days whose baseline day (7 days earlier) was a holiday.
Output	`holiday_impact.csv`: group (holiday, week-after-holiday, regular), WAPE, Bias %, sales.
Insight to look for	A clear rule: the baseline breaks in two places, on the holiday and one week later.

Q8. What is the effect of promotions?	
Business question	Does error rise when items are on promotion?
Hypothesis	Promotion days have higher WAPE and lean to shortfall, because sales jump above last week.
Method	Group rows by onpromotion = 0 and onpromotion above 0. WAPE, Bias % by family group.
Output	`promotion_impact.csv`: promo group, WAPE, Bias %, shortfall units.
Insight to look for	Whether promotions should feed the forecast as an input.

Q9. Is there a payday effect?	
Business question	Is error higher around the 15th and the last day of the month?
Hypothesis	Paydays raise sales and the baseline under-forecasts them.
Method	Flag the 15th and last day (and one day after). WAPE and Bias % for flagged vs other days.
Output	`payday_impact.csv`: payday group, WAPE, Bias %, sales.
Insight to look for	A simple calendar rule that the planners can use.

Q10. Which weekdays are hardest?	
Business question	Does error change by day of the week?
Hypothesis	Weekend days have more error than weekdays.
Method	`EXTRACT(DOW FROM date)`. WAPE and Bias % by weekday and by family group.
Output	`weekday_error.csv`: weekday, WAPE, Bias %, sales.
Insight to look for	Whether the stock plan needs a weekday adjustment.

Q11. Which baseline is better for which family?	
Business question	Is Baseline B (4-week average) better than Baseline A (last week) for some families?
Hypothesis	B is better for smooth families. A is better for families that move fast.
Method	WAPE for A and B by family. Compute the difference. Label the better one.
Output	`baseline_compare.csv`: family, WAPE A, WAPE B, better baseline.
Insight to look for	A practical tip: use B for volatile families. This is a simple comparison, not model building.

Q12. Is the error getting better or worse?	
Business question	Does WAPE trend up or down over time for the top families?
Hypothesis	Error is stable over time, except in event months.
Method	WAPE by month and year for the top 5 families. Use window functions for a 3-month rolling WAPE.
Output	`error_trend.csv`: year-month, family, WAPE, rolling WAPE.
Insight to look for	A stable problem needs a process fix. A growing problem needs urgent action.

4. SQL skills shown by this project

Technique	Used in
JOINs (inner, left)	Q5, Q7 and the holiday flag table
CTEs	Baseline, error table, Q3, Q11
Window functions: LAG, AVG OVER, SUM OVER, RANK	Baseline, Q2, Q3, Q5, Q12
CASE WHEN	Direction labels, periods, flags
Date functions: EXTRACT, DATE_TRUNC	Q6, Q9, Q10, Q12
Views and clear script order	All steps
Data quality checks	Step 03 and the QA list

5. Output files for the report and checks

These CSV files go into `01_Datasets\processed`. The dashboard uses the error fact table (so slicers work together). The CSV outputs are used for the report and to **cross-check dashboard totals**.

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Power BI Dashboard Requirements

> **In simple words**
>
> This is the blueprint for the dashboard. It is built as a story: each page has a headline that states the insight, and the last page gives the actions.

1. Design idea: the dashboard tells a story

A manager should see the main message in **30 seconds**. So every page has a **headline title written as a sentence** (not just a chart name), a few visuals, and one text box with the takeaway.

Weak chart title	Insight headline (better)
WAPE by Family	[X] families cause [Y]% of total forecast error
Bias by Month	[Month] is the only month where we clearly under-forecast
WAPE by City	Error is high in [N] small cities, not everywhere

The headline can be a **dynamic measure** (a text measure that fills the family name and numbers from the data), so it updates with filters.

2. Data model

- Import the **error fact table** (store x family x date) from PostgreSQL. Add dimension tables: date, store, family.
- Relationships: fact to date, store, family. One direction, one-to-many.
- If the file is too slow, aggregate to store x family x week in SQL and say so in the report.
- Rename columns to readable names (Store Number, not store_nbr). Hide key columns.

3. Global slicers (every page)

Slicer	Field	Style
Date	Year and month	Dropdown or range
Region	State, then city	Hierarchy dropdown
Family	Product family	Dropdown with search
Store type	Type A to E	Buttons

4. Pages

Page 1: Executive Summary	
Headline	Overall accuracy is [X]%. [N] families cause [Y]% of the error.
Manager question	Are we doing well overall? Where is the problem?
Visuals	KPI cards: WAPE, Accuracy %, Bias %, Excess units, Shortfall units. Bar: top 5 families by WAPE. Line: monthly WAPE trend. Text box: 3 key takeaways.
Interaction	Click a family to filter the page. Button to Page 5.

Page 2: Category Deep-Dive	
Headline	[Family list] are the priority: high error and high volume.
Manager question	Which families should we fix first, and in which direction do they miss?
Visuals	Scatter: x = total sales, y = WAPE, one dot per family (top right = priority). Pareto bar + cumulative line. Stacked bar: excess vs shortfall units by family.
Interaction	Drill-through to Family Detail page (daily actual vs forecast).

Page 3: Regional Analysis	
Headline	Error is highest in [states or cities], mostly where volume is [low/high].
Manager question	Is the problem everywhere or in specific places?
Visuals	Map or bar: WAPE by city. Matrix: state > city with WAPE, Bias % and conditional colors. Bar: WAPE by store type.
Interaction	Tooltip: top error family in that city.

Page 4: Seasons and Events	
Headline	[Month] and [event] are the hardest periods. Holiday weeks lean to [excess/shortfall].
Manager question	When does the baseline fail, and which way?
Visuals	Column: WAPE by month. Bar: holiday vs week-after-holiday vs regular. Matrix heatmap: weekday x month. Cards: payday vs other days, promotion vs no promotion.
Interaction	Slicer for named period (Christmas, Carnival, school start).

Page 5: Insights and Actions	
Headline	Five findings and what we recommend.
Manager question	What should we do, who owns it, and how soon?
Visuals	5 insight cards (text boxes): finding, evidence number, recommendation, owner, priority (High, Medium, Low). No new charts. Link each card to the page that proves it.
Interaction	Buttons back to pages 1 to 4.

5. Hidden drill-through page: Family Detail

Line chart of daily actual vs forecast for one family, with holiday days marked. Cards for WAPE, Bias % and top 3 error days. Helps explain why a family is high error.

6. Measures to create (names and logic)

Measure	Logic (plain words)
Total Actual Sales	Sum of actual sales
Total Forecast	Sum of forecast sales
Total Abs Error	Sum of absolute error
WAPE %	DIVIDE(Total Abs Error, Total Actual Sales)
Accuracy %	1 - WAPE %
Bias %	DIVIDE(Total Forecast - Total Actual Sales, Total Actual Sales)
Excess Units	Sum of excess_units
Shortfall Units	Sum of shortfall_units
Error Share %	Abs error of selected group / abs error of all groups
High Error Flag	1 if WAPE % above 30% and sales above the minimum volume
Direction Label	Over if Bias % above +5%, Under if below -5%, else Balanced
Headline Text	Text measure that puts the top family and its numbers into a sentence

Use `DIVIDE` so a zero in the bottom does not crash the measure. Never average a WAPE column. Always use the measures.

7. Design rules

- Maximum 6 visuals per page. One message per visual.
- Colors: gray-blue by default. Red only for high error. Orange for excess, blue for shortfall, and use them the same on every page.
- No 3D, no pie with more than 4 slices, sort bars from high to low.
- Axis and card titles in plain words, units shown (units, %).
- Add a **Reset filters** bookmark and page navigation buttons.
- A small note on each page: data period and baseline used.

8. Dashboard checks before sharing

- Totals on Page 1 match the SQL totals (sales, abs error, WAPE).
- Every slicer changes the numbers it should.
- Every headline is true for the current filters (or says 'all data').
- No visual is empty, cut off or overlapping on the screen.

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Python Quick Charts (Optional)

> **In simple words**
>
> Python is optional here. We use it only for a few quick charts. All real work is in SQL and Power BI.

> **This document is optional**
>
> Use Python only for 2 or 3 quick charts and to look at the data once. Cleaning, baseline and analysis are done in SQL. Do not build models.

1. What Python is for

Use	Why it helps
Line chart of daily sales for 3 families	See seasonality and spikes before writing SQL
Histogram of sales for one family	See if many days are zero
Heatmap of WAPE (weekday x month) from SQL output	A nice image for the report

2. What not to do

- No forecasting models (ARIMA, Prophet, XGBoost, LSTM).
- No feature engineering or tuning.
- No cleaning that is not also done in SQL. The SQL scripts are the source of truth.

3. Libraries

- **pandas**: read the SQL output CSV files.
- **matplotlib / seaborn**: simple charts.

4. Output

One notebook `eda_charts.ipynb` in `06_Python_Optional` and 2 or 3 PNG images in `07_Screenshots`.

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Report Writing Guidelines

> **In simple words**
>
> This document explains how to write the final report so that each page gives a clear finding and a clear action.

1. Report structure (target: 8 to 10 pages)

Section	What to write	Length
1. Executive Summary	Question, 3 main findings, 3 recommendations. Use the template in Doc 13.	1 page
2. Business Context	Problem, decisions, stakeholders	half page
3. Data and Method	Data used, baseline rule, metrics, high-error rule, limits	1 page
4. Findings	One short section per insight (use the insight card from Doc 12)	3 to 4 pages
5. Recommendations	Action table: action, owner, priority, evidence, how to measure	1 page
6. Limitations and Next Steps	What the data cannot tell, what to do next	half page
Appendix	Metric definitions, SQL file list, data checks	1 page

2. How to write a finding

1. **Headline:** the finding in one sentence.
2. **Evidence:** one chart and the key numbers.
3. **Compare:** against overall, another group or another period.
4. **Reason:** the likely cause. Say 'likely' unless the data proves it.
5. **Action:** what to do next.

3. Recommendation table (example format)

Action	Owner	Priority	Evidence	How to measure
[What to change]	[Team]	High / Medium / Low	[Finding and number]	[WAPE or Bias after change]

4. Writing rules

- Lead with the answer. Details come after.
- Use simple words. Explain a term the first time.
- Every number has a comparison (higher than what?).
- One message per chart. Put the message in the chart title.
- Use 'is linked to' or 'is likely caused by', not 'is caused by', unless proven.
- No invented numbers. If cost data is missing, say so and use units.

5. Presentation

Slide plan, speaker notes and the 2-minute story are in **Doc 13**.

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Project Plan and QA Checklist

> **In simple words**
>
> This is the timeline and quality check. It includes checks for the insights, not only for the numbers.

1. Timeline (7 working days)

Day	Goal	Output
1	Charter, requirements, folders, load data	Docs 01-02 done, tables in PostgreSQL
2	Data checks, holiday flag table, baseline	Scripts 03 to 05
3	Error table and Q1 to Q6	Fact table, first outputs, insight log started
4	Q7 to Q12, finish insight log	All CSV outputs
5	Power BI model and Pages 1 to 3	Working dashboard draft
6	Pages 4 and 5, checks, screenshots	Final .pbix, checked totals
7	Report, executive summary, slides, GitHub	Report, deck, README

2. QA checklist (tick before sharing)

Area	Done	Check	How
Data	[ ]	Row counts match Kaggle; 54 stores and 33 families found	COUNT checks
Data	[ ]	Missing dates are known (for example 25 Dec) and not filled	Date gap query
Data	[ ]	Rows before store start date are excluded	Compare counts before and after
Data	[ ]	Holiday flag table has one row per store and date	COUNT vs COUNT DISTINCT
SQL	[ ]	Baseline uses the exact earlier date (7, 14, 21, 28 days)	Spot-check 5 rows by hand
SQL	[ ]	Total sales in error table = total sales in train after exclusions	SUM compare
SQL	[ ]	WAPE is computed from sums, never averaged	Read the query
SQL	[ ]	Scripts are numbered, commented, and run from top to bottom	Run all in order
Metrics	[ ]	WAPE for one family matches a manual calculation	Hand calculation
Metrics	[ ]	Excess units - shortfall units = total forecast - total actual	Identity check
Dashboard	[ ]	Page 1 totals match SQL totals	Compare cards to SQL output
Dashboard	[ ]	Every slicer changes the numbers	Click test
Dashboard	[ ]	Every headline is true for the current filters	Read each page
Dashboard	[ ]	No overlapping or cut-off visuals	View in full screen
Insights	[ ]	Every insight has evidence, a comparison, a reason and an action	Check the insight log
Insights	[ ]	Causes are written as 'likely' unless proven	Read the report
Insights	[ ]	No invented numbers; impact is in units	Read the report

Area	Done	Check	How
Insights	[ ]	Limits are stated (baseline is simple, no cost data)	Read the summary
Portfolio	[ ]	README has problem, method, insights, screenshots	Open GitHub page
Portfolio	[ ]	SQL scripts, report and slides are uploaded	Check the repo

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Insights Analyst Guide

> **In simple words**
>
> This guide shows how an Insights Analyst thinks and works: from data to information to insight to action. It also gives the step roadmap with purpose and result.

1. Your lane: Insights Analyst

Companies now split analyst work into two lanes. **Insights Analysts** talk to the business, tell stories with data and drive decisions. **Analytics Engineers** build pipelines and production-level data models. You chose the first lane. Your projects, resume and learning should all point there.

	Insights Analyst (you)	Analytics Engineer (not now)
Main question	What does this mean and what should we do?	How do we make the data reliable and automatic?
Talks to	Managers and business teams	Data engineers and analysts
Tools	SQL, Power BI, Excel, storytelling	dbt, Airflow, cloud, pipelines
Output	Insights, dashboards, recommendations	Clean models and pipelines

2. From data to action (the ladder)

Level	Example format	Who stops here
Data	Dairy sales: 1,200 units on Monday	Beginners
Information	Dairy WAPE is [X]%	Junior analysts
Insight	Dairy WAPE is [X] points above overall, mostly after holidays	Good analysts
Action	Do not copy last week's holiday sales into next week's order	Insights analysts

Numbers in this table are placeholders. Use your real results.

3. Five-step habit

1. **Ask:** What decision does this support?
2. **Check:** Is the data clean and complete?
3. **Analyze:** Compare groups and periods.
4. **Explain:** Why? Test a hypothesis. Say 'likely' if unproven.
5. **Recommend:** Action, owner, priority, how to measure.

4. Skills map

Need now	Good to have	Skip for now
SQL (joins, CTEs, window functions)	Basic statistics (M.Sc Maths helps)	Airflow, dbt, Spark
Power BI and DAX basics	Excel pivots and lookups	Cloud data platforms
Business storytelling	A/B testing basics	Machine learning models
Stakeholder communication	Python for charts	Production data pipelines

5. Roadmap: purpose and result of each step

Step	Purpose	Result
1 Folders and tools	Keep work organized	Folder tree, PostgreSQL ready
2 Load data	Get data into SQL	5 tables with correct row counts
3 Data checks	Find problems early	List of issues and how they are handled
4 Holiday flag table	Avoid double counting	One row per store and date
5 Baseline forecast	Create a guess to compare	Baseline A and B in a view
6 Error table	Measure the miss	Fact table with error, excess, shortfall, flags
7 Analysis Q1 to Q12	Find where, when, which way	CSV outputs and an insight log
8 Power BI model	Make exploration easy	Star model, measures that match SQL
9 Dashboard pages	Show the story	5 pages with headline titles
10 Insights and actions	Turn findings into advice	6+ insight cards, action table
11 Report	Explain to managers	8 to 10 page report
12 Presentation	Persuade	8-slide deck and 2-minute story
13 Portfolio	Be found by recruiters	GitHub repo, README, resume lines

6. Common mistakes

- Showing numbers without a comparison or an action.
- Presenting a cause as a fact when it is only a guess.
- Inventing money values when the data has no cost.
- Too many charts. Fewer, clearer charts win.
- Using jargon with managers.

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Fast Track Steps and Flow

> **In simple words**
>
> This is the short action list. Do one step at a time, tick it, and move on.

1. Flow



2. Step checklist

#	Step	Time	What to do	Result
1	START HERE: Folders and PostgreSQL	1h	Create the folder tree. Install and open PostgreSQL. Create a database `demand_forecasting`.	Empty database, folders ready
2	Load data	3h	Create tables for the 5 CSVs. Load them. Compare row counts with Kaggle.	5 tables loaded
3	Data checks	2h	NULLs, duplicates, missing dates, store first sale dates. Write issues in a notes file.	Issue list
4	Holiday flag table	3h	Filter transferred = False. Match National, Regional, Local. One row per store and date.	Flag table without duplicates
5	Baseline A and B	4h	Use LAG or a self-join on the exact date (7, 14, 21, 28 days before).	Baseline view
6	Error table	3h	Join actual, baseline, flags. Add abs error, excess, shortfall. Exclude rows before store start.	Fact table, totals checked
7	Analysis Q1 to Q6	5h	Write one query per question. Export CSV. Add a row to the insight log each time.	6 outputs, 6 log rows
8	Analysis Q7 to Q12	5h	Events, promotions, payday, weekday, baseline compare, trend.	6 more outputs
9	Power BI model	4h	Import fact and dimensions. Create measures. Check totals against SQL.	Model with matching totals
10	Dashboard pages 1 to 5	7h	Build with headline titles. Page 5 uses your insight cards.	Final .pbix
11	Report	5h	Follow Doc 08. Use the insight cards.	8 to 10 page report
12	Presentation	3h	Follow Doc 13. Practice the 2-minute story.	8 slides
13	GitHub and resume	3h	Upload scripts, report, slides, screenshots. Write README. Add 3 resume lines.	Public repo

3. Minimum version if you run short of time

- Baseline A only. Analysis Q1 to Q6. Dashboard pages 1, 2 and 5.
- Report with only 4 findings. Slides: summary, 3 findings, actions.

- Never skip: data checks, matching totals, recommendations, README.

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Hypothesis and Insight Playbook

> **In simple words**
>
> This document teaches how to turn SQL numbers into insights and actions. It gives the sentence formula, the insight card, a list of hypotheses to test, and a log to fill in.

1. What is an insight?

An **insight** is a finding that explains something and leads to action. It answers **What? Compared with what? Why? So what?**

Not an insight	An insight
WAPE for the family is [X]%.	WAPE for the family is [X] points higher than overall, and most of it comes from the week after holidays. Do not copy holiday sales into next week's plan.
December has more sales.	In December the baseline is [too low / too high] by [X]%, so plan buffer stock for [family].

Anything in [brackets] is a placeholder. Replace it with your real result.

2. Insight sentence formula

> **Use this every time**
>
> **[Where / when]** has **[metric] of [X]**, which is **[higher / lower] than [comparison]**. This is **likely because [reason]**. We recommend **[action]**, owned by **[team]**.

3. Insight card (one per finding)

Field	Write here
ID and title	I-01: short headline
Question answered	Which business question from Doc 01 (A to E)
Evidence	Key numbers and the chart / query that shows it
Comparison	Compared with overall, another group, or another period
Likely reason	Hypothesis result: confirmed, likely, or needs more data
Recommendation	Specific action
Owner and priority	Team and High / Medium / Low
Expected effect	In units or WAPE points, not money
Limit	What this does not prove

4. Hypotheses to test

These are starting guesses, **not facts**. Test each with the SQL question shown. A rejected hypothesis is still a useful finding.

ID	Hypothesis	Why we think so	Test	If true, possible action
H1	Error is higher on holidays	Demand jumps and last week is normal	Q7	Holiday-specific forecast or buffer
H2	One week after a holiday the baseline is too high	It copies the holiday spike	Q7	Do not use holiday weeks as the base
H3	December is the hardest month	Christmas shopping	Q6	Plan December separately
H4	Low-volume families have the highest WAPE	Small numbers are noisy	Q2	Track them, do not over-fix
H5	A few families cause most of the error	Big families dominate units	Q3	Fix the top families first
H6	Promotions raise error and cause shortfall	Sales jump above last week	Q8	Feed promotion plan into the forecast
H7	Paydays (15th and last day) raise sales above baseline	Wages paid on those days	Q9	Payday calendar adjustment
H8	Weekends have more error	Shopping patterns differ	Q10	Weekday adjustment
H9	Smaller cities and some store types have higher WAPE	Lower volume, more noise	Q5	Different rules by store type
H10	A 4-week average beats last week for volatile families	Averaging reduces noise	Q11	Use the better baseline by family

5. Confidence labels

Label	Meaning
Confirmed by data	Clear pattern, large size, holds across groups
Likely	Pattern is clear, but the cause is our guess
Needs more data	Small groups or mixed results

Correlation is not causation. A pattern shows that two things move together. It does not prove one causes the other.

6. Types of recommendation

Type	Use when	Example form
Adjust the method	A simple baseline fails in a clear way	Use a 4-week average for [family]
Add buffer stock	Shortfall units are high	Add [X]% buffer for [family] in [month]
Reduce orders	Excess units are high	Lower the order for [family] in [region]
Plan events	Holidays, paydays, promotions break the pattern	Add an event calendar to planning
Monitor	Evidence is weak or volume is tiny	Track weekly, review next quarter
Fix data	Data gaps cause error	Record store opening dates

Rank actions by **impact** (share of error or units) and **effort** (easy or hard). Do high impact and low effort first.

7. Insight log (fill in as you analyze)

ID	Question / SQL	Finding with numbers	Reason	Action	Confidence
I-01					
I-02					
I-03					
I-04					
I-05					
I-06					
I-07					
I-08					

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Executive Summary and Presentation Template

> **In simple words**
>
> This document gives fill-in templates for the executive summary, the slides, the dashboard walk-through and the questions you will be asked.

1. One-page executive summary (template)

Part	Write here
Question	In one sentence: where is demand hard to predict, and what should we do?
Answer in one line	[N] families cause [Y]% of forecast error. Errors rise in [period/event]. We recommend [main action].
Finding 1	[Headline + number + comparison]
Finding 2	[Headline + number + comparison]
Finding 3	[Headline + number + comparison]
Recommendation 1	[Action, owner, priority]
Recommendation 2	[Action, owner, priority]
Recommendation 3	[Action, owner, priority]
What we cannot say	[Baseline is simple; no cost data; reasons are likely, not proven]

2. Story structure

Part	Meaning	Your project
Situation	What is normal	Stores need to forecast demand every day
Complication	What goes wrong	Forecasts miss in certain places and times
Question	What we asked	Where, when and which way do we miss?
Answer	What we found and recommend	Top families, periods, direction, actions

3. Slide plan (8 slides)

#	Slide	Headline (write as a sentence)	Visual
1	Title	Where demand is hard to predict: a forecast error study	Title and project name
2	Why it matters	Wrong forecasts lead to excess stock or empty shelves	One simple graphic
3	The big picture	Overall accuracy is [X]%, with a bias of [Y]%	KPI cards
4	Which families	[N] families cause [Y]% of the error	Pareto chart
5	Which way	[Family group] over-forecast, [group] under-forecast	Excess vs shortfall bars
6	Where and when	Error is highest in [places] and during [periods]	Region chart and month chart
7	Why (likely)	The baseline breaks on [holidays, promotions, paydays]	Holiday vs regular chart
8	What to do	Three actions, with owners and priority	Action table

4. Dashboard walk-through script

Time	What to say
30 seconds	Page 1 only: overall accuracy, top families, the main message.
2 minutes	Pages 1, 2 and 4: which families, which way, when it breaks.
5 minutes	All pages, then Page 5 actions. Invite questions.

5. Questions you will be asked

Question	Good answer approach
Why did you use a simple baseline?	The data has no company forecast. A baseline shows where demand is hard to predict, and anyone can understand it.
Can I trust these numbers?	Name the checks: totals match SQL, holiday join tested, missing dates handled.
Why WAPE and not MAPE?	Many rows have zero sales. MAPE breaks on zero. WAPE does not.
What would it be worth in money?	The data has no price or cost. I measured units. With cost data I could convert.
What would you do next?	Try the better baseline per family, add promotion plans, track weekly.

6. GitHub README outline

1. Title and one-line summary
2. Business problem and decisions
3. Data and tools (PostgreSQL, Power BI)
4. Method (baseline, metrics)
5. Top 5 insights with screenshots
6. Recommendations
7. Limits
8. How to run the SQL scripts

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Interview and Portfolio Guide

> **In simple words**
>
> This document helps you present the project in interviews, on your resume and on GitHub. It has a 2-minute script, 15 questions with answer approaches, and resume lines.

1. Your project in 2 minutes (fill the blanks)

> **Script**
>
> I worked on a demand forecasting study for a grocery retail case. The business problem was that planners did not know where demand was hard to predict. I built a simple baseline in SQL, measured error using WAPE and Bias, and analysed it by product family, region, season and events. I found that **[N] families caused [Y]% of the error**, and that error rose around **[event or period]**. I built a 5-page Power BI dashboard and recommended **[main action]**. I also stated the limits: it is a simple baseline and the data has no cost, so I measured impact in units.

2. Interview questions and how to answer

Question	Answer approach
Tell me about this project	Use the 2-minute script.
What was the business problem?	Decision problem: where to add stock, where to cut orders.
How did you build the baseline?	Same weekday last week, plus a 4-week average. Window functions with exact date checks.
Why WAPE?	Works with zero sales, weights big sellers, easy to explain.
What is Bias and why show it?	Direction of the miss. WAPE alone hides if we are too high or too low.
How did you avoid double counting holidays?	Store-date flag table. Match National, Regional, Local. Remove transferred holidays.
How do you know your numbers are right?	Totals match across SQL and Power BI. Hand-checked one family.
What was the most important insight?	Your top Pareto finding and the action.
Did you find anything that surprised you?	A rejected hypothesis is a good answer.
How did you design the dashboard?	Story-first: headline titles, one message per chart, actions on the last page.
Which SQL techniques did you use?	CTEs, window functions, joins, CASE, date functions, views.
What are the limits?	Simple baseline, no cost, causes are likely not proven.
Is this a data scientist project?	No. I did not build models. I measured and explained where simple forecasts fail, and gave actions.
What would you do with more time?	Add cost data, event calendars, better baselines by family.
How would you explain WAPE to a manager?	Out of every 100 units sold, about [X] are the size of our miss.

3. Resume lines (replace brackets with your real results)

- Analysed [N] million daily sales rows in PostgreSQL to measure forecast error (WAPE, Bias) by family, region and season.
- Identified that [N] product families drive [Y]% of forecast error and recommended [action].

- Built a 5-page Power BI dashboard with headline insights and an actions page for supply chain managers.
- Wrote 12 SQL analyses with CTEs and window functions, and validated totals between SQL and Power BI.

Add a number only after you have it from your own results. Do not copy the brackets.

4. LinkedIn headline ideas

- Insights Analyst | SQL, Power BI, Excel | Turning data into business decisions
- M.Sc Mathematics | Aspiring Insights Analyst | SQL and Power BI projects

5. Portfolio plan

Project	Tools	What it shows
HR Job Change (Excel)	Excel	Cleaning, pivots, dashboard, business story
Demand Forecasting (this one)	PostgreSQL, Power BI	Advanced SQL, metrics, insights, actions
EduPulse	Power BI	Retention dashboard and recommendations

6. Red flags to avoid

- Calling the project 'forecasting model' or 'machine learning'. It is analysis.
- Showing a dashboard without saying what to do.
- Quoting numbers you cannot explain.
- Claiming money savings when there is no cost data.

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